

Smart Autonomous Delivery Robot

for Campus Navigation

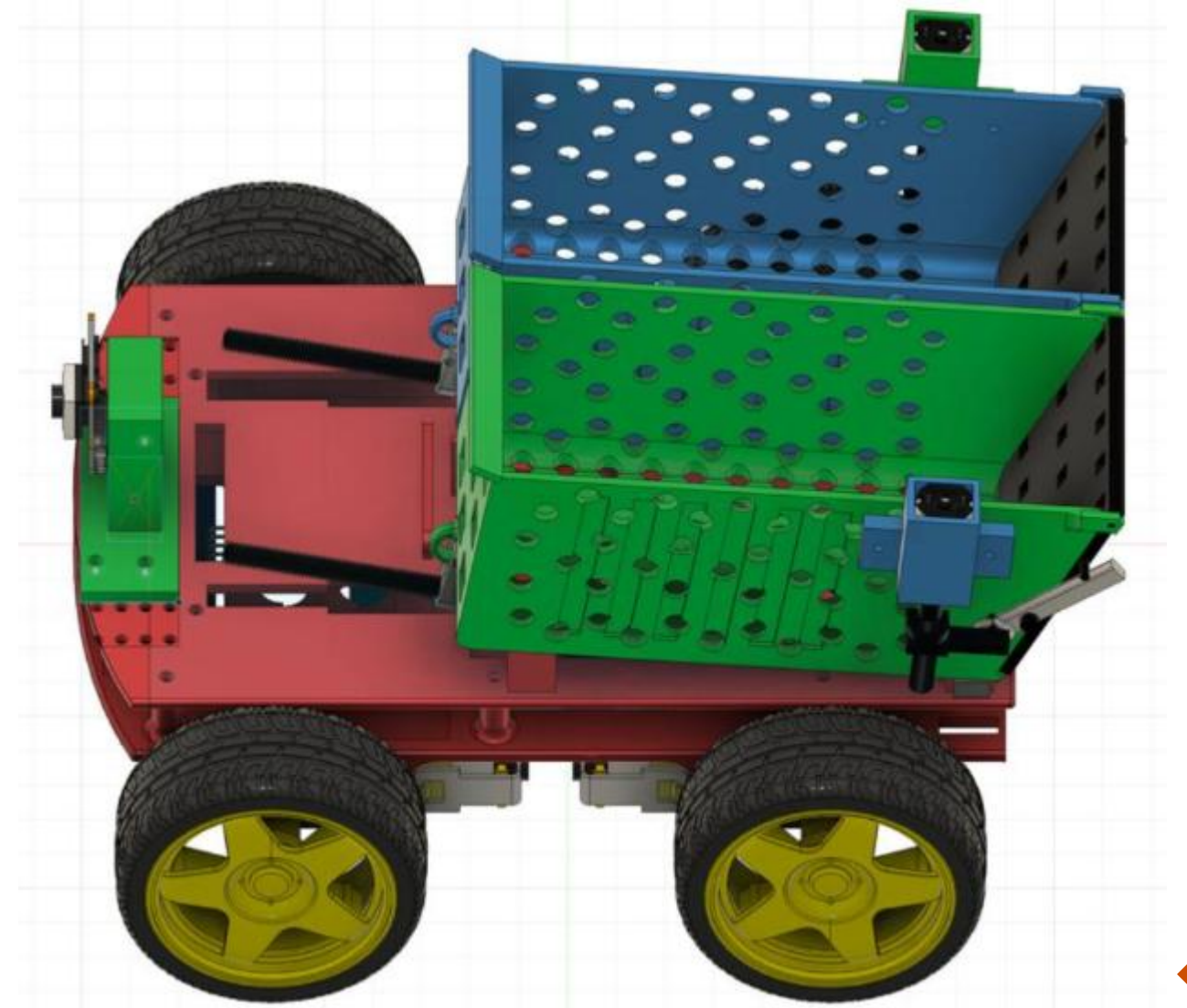
Team RoboxCraft · Smart Mobility & Transportation

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PROBLEM DEFINITION

In large campuses, delivering small items like documents, medicines, or packages between departments is time-consuming and inefficient. Manual delivery leads to delays and requires continuous human effort.

Functional Requirements

1. Path Following

Autonomous line detection using IR sensors

2. Stop Detection

Identify delivery points via visual markers (cards)

3. Delivery Execution

Stop and deliver items using servo mechanism

4. Path Recovery

Auto-return to path after each delivery

Constraints

1. Indoor Only

No GPS — designed for controlled indoor spaces

2. 14-Hour Build

Must be simple and reliable within hackathon time

3. Basic Sensors

Only IR sensors and MU Vision module available

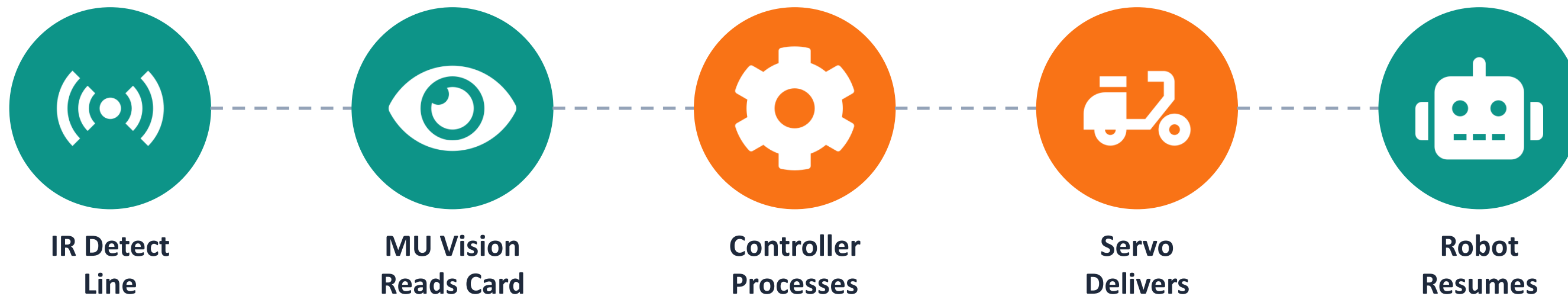
4. Low-Cost

Built with affordable, off-the-shelf components

PROPOSED SOLUTION

A smart autonomous robot that follows a predefined path via IR sensors, detects delivery stops using MU Vision Sensor, performs item drop via servo, then automatically resumes navigation, zero human intervention required.

HOW IT WORKS



CORE CAPABILITIES

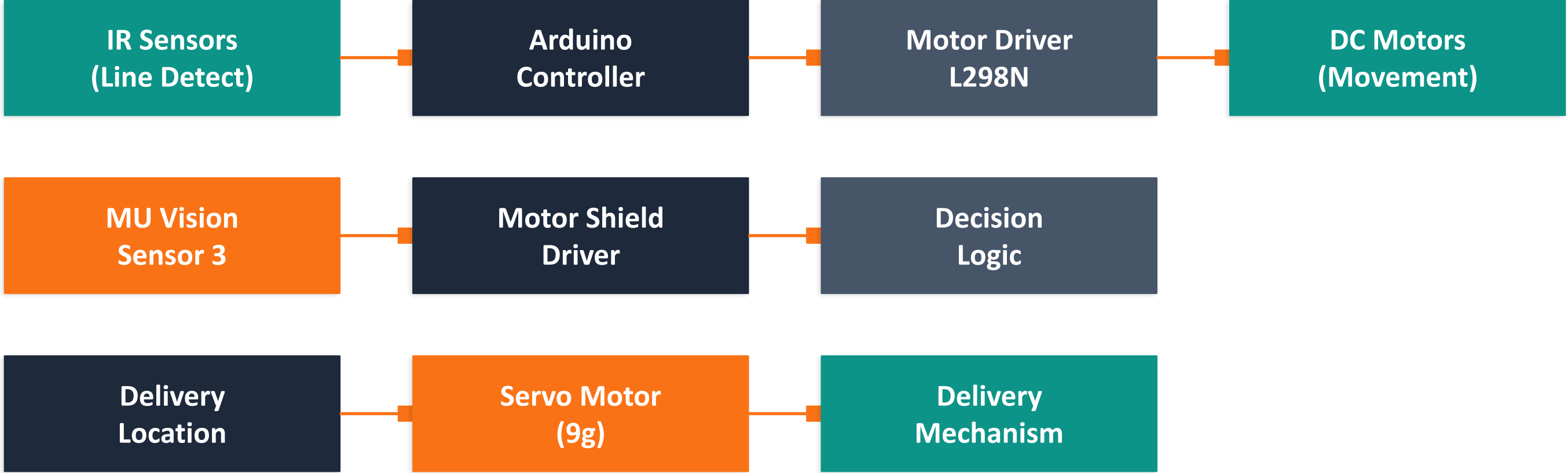
Autonomous Line Following

Vision-Based Stop Detection

Servo-Based Item Delivery

Automatic Path Recovery

SYSTEM ARCHITECTURE



COMPONENTS

- ▶ Arduino Uno / Nano
- ▶ Advanced IR Sensors (Line detection)
- ▶ MU Vision Sensor 3
- ▶ L298N Motor Driver Shield
- ▶ DC Motors + Chassis
- ▶ Servo Motor (Delivery)
- ▶ Battery Pack 7.4V
- ▶ Jumper Wires + Switch

CONNECTION & FEASIBILITY

IR Sensors (Line Detection)

IR1 → A0
IR2 → A1
IR3 → A2
IR4 → A3
IR5 → A4
VCC → 5V
GND → GND

MU Vision Sensor 3 (Card Detect)

- ▶ VCC → 5V
- ▶ GND → GND
- ▶ TX → Arduino D2
- ▶ RX → Arduino A5

Motor Driver L298N

- ▶ IN1, IN2 → Direction pins
- ▶ IN3, IN4 → Direction pins
- ▶ ENA, ENB → PWM speed
- ▶ OUT1–4 → Left & Right motors

Servo Motor (Delivery)

- ▶ VCC → 5V
- ▶ GND → GND
- ▶ Signal → PWM pin D10
- ▶ Battery → Motor driver input

EXCEPTED OUTCOME

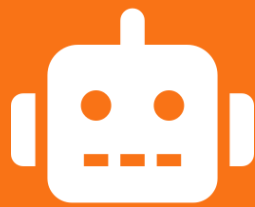
WORKING MODEL

A fully functional autonomous delivery robot that:

- Follows a predefined track using IR sensors
- Detects delivery stations using vision card recognition
- Drops items via servo mechanism at correct stops
- Automatically returns to path and resumes navigation



Saves time on
campus deliveries



Reduces manual
human effort



Demonstrates
practical automation



Scalable to real
smart campus use

TECH STACK

Controller

Arduino Uno / Nano

Path Sensing

2× IR Sensors

Vision

MU Vision Sensor 3

Motor Control

L298N Driver

Actuation

Servo Motor (9g)

Drive

DC Motors + Chassis

Power

7.4V Battery Pack